

Pass Decision Making in Soccer: An MDP Approach

Extending Van Roy et al.'s Leaving Goals on the Pitch: Evaluating Decision Making in Soccer

Martim Rêgo

Deep Learning & AI in Sport

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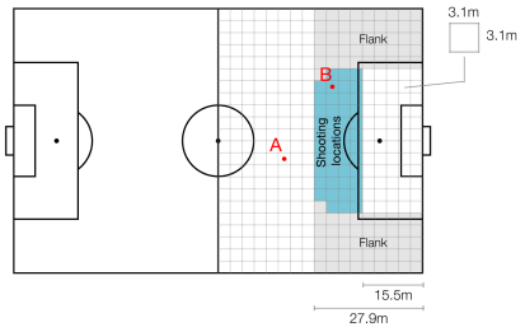
Van Roy et al. (2020): "Leaving Goals on the Pitch"

Setup:

- EPL 2017-19
- **Question:** Shoot from distance or pass?
- 22×17 grid (374 states)
- Actions: Move or shoot

MDP Components:

- **States:** Field zones
- **Actions:** Move/shoot
- **Transitions:** $P(s'|s, a)$
- **Policy:** $\pi(a|s)$
- **Rewards:** Goals



Key Findings:

- Teams shoot **too rarely** from distance
- 10-20% more long shots $\rightarrow$ +1 goal/season
- Specific zones: shooting $>$ passing

Our Extension: Pass Decision Analysis

What We Changed:

- ✓ Action space: 8 actions
 - 6 pass types (length $\times$ direction)
 - Shoot
 - Carry (dribbling)
- ✓ Grid: 7×11 (77 states) — simplified
- ✓ Data: Premier League 2024/25 (521K actions)
- ✓ Action masking for physical constraints

Pass Classification:

- **Length:** Short ($\leq 25\text{m}$) vs Long ($> 25\text{m}$)
- **Direction:** Forward / Lateral / Backward
 - Angle-based

Action Masking:

- Shooting: disabled $> 30\text{m}$ from goal
- Long forward: disabled near goal
- Long backward: disabled near own goal

Data Volume:

- 378 matches (PL 2024/25 season)
- 521,470 total actions
 - 328,713 passes (63.0%)
 - 184,080 carries (35.3%)
 - 8,677 shots (1.7%)
- 20 teams analyzed

Success Rates:

- Passes: 82.85%
- Shots (goals): 12.3%
- Carries: 91.6%

State-Action Coverage:

- 579 unique (s, a) pairs observed
- 94% coverage (excellent!)
- Mean: 900.6 obs/pair

For Each Team:

① Transition Matrix P

- Shape: (80, 8, 80)
- 77 field + 3 absorbing states
- **Learned from data:**
 - Passes/carries: Laplace smoothing ($\alpha = 2.0$)
 - Shots: Bayesian shrinkage ($\alpha = 10$)

② Policy Matrix π

- Shape: (77, 8)
- Observed action frequencies

③ Reward Vector R

- $R(\text{goal}) = 1$, else 0

Absorbing States:

- State 77: Goal (successful shot)
- State 78: No goal (failed shot)
- State 79: Loss of possession

Position-Based xG :

- Geometric model: viewing angle to goal
- $xG = 0.76 \times \left(\frac{\theta}{36.8^\circ}\right)^{1.5}$
- Used for shot transitions

Fundamental Matrix: Theoretical Foundation

Definition:

$$\mathbf{N} = (\mathbf{I} - \mathbf{Q})^{-1}$$

where $\mathbf{Q}$ contains transition probabilities between transient (field) states.

Interpretation:

- N_{ij} = expected visits to state j from state i
- Captures **all possible paths** before absorption
- Multi-step sequences handled automatically

Computing Expected Goals:

$$E[\text{goals}|s_0] = \sum_j N_{s_0,j} \cdot \pi(\text{shoot}|j) \cdot xG(j)$$

Why It's Powerful:

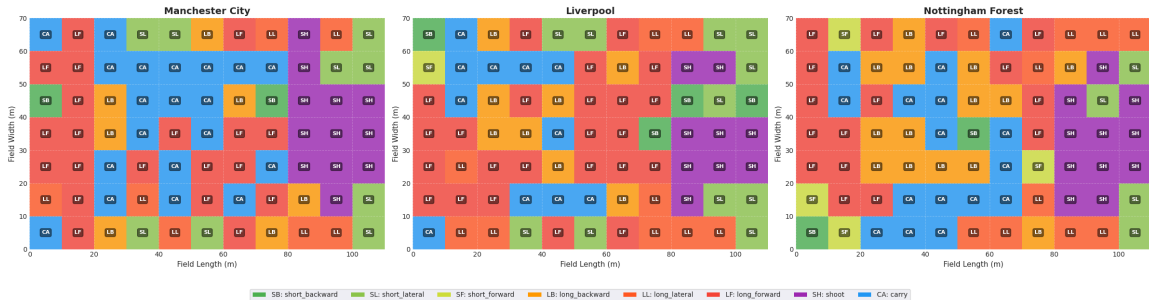
- Exact calculation (no simulation)
- Handles uncertainty
- Enables counterfactual reasoning
- Foundation for policy evaluation

Computed Expected Goals per Team for the whole season

Liverpool	:	159.42	xG from 5,168 possessions (0.0308 per poss)
Arsenal	:	121.59	xG from 4,471 possessions (0.0272 per poss)
Manchester U	:	104.44	xG from 5,080 possessions (0.0206 per poss)
Newcastle	:	129.45	xG from 4,918 possessions (0.0263 per poss)
West Ham	:	109.60	xG from 4,926 possessions (0.0222 per poss)
Aston Villa	:	122.94	xG from 4,462 possessions (0.0276 per poss)
Manchester City	:	133.24	xG from 4,528 possessions (0.0294 per poss)
Everton	:	109.34	xG from 4,965 possessions (0.0220 per poss)
Tottenham	:	125.41	xG from 5,014 possessions (0.0250 per poss)
Fulham	:	117.04	xG from 5,049 possessions (0.0232 per poss)
Chelsea	:	129.72	xG from 4,879 possessions (0.0266 per poss)
Wolverhampton	:	128.85	xG from 4,924 possessions (0.0262 per poss)
Southampton	:	87.72	xG from 4,486 possessions (0.0196 per poss)
Crystal Palace	:	107.33	xG from 5,138 possessions (0.0209 per poss)
Leicester	:	90.15	xG from 4,752 possessions (0.0190 per poss)
Bournemouth	:	112.21	xG from 5,064 possessions (0.0222 per poss)
Brighton	:	134.67	xG from 5,061 possessions (0.0266 per poss)
Nottingham	:	130.78	xG from 4,760 possessions (0.0275 per poss)
Ipswich Town	:	91.79	xG from 4,554 possessions (0.0202 per poss)
Brentford FC	:	142.29	xG from 5,101 possessions (0.0279 per poss)

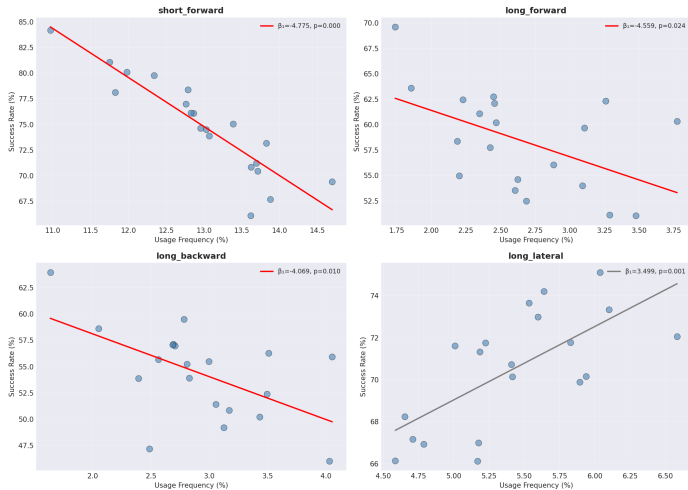
Result 1: Optimal Action Selection

Research Question: Which action maximizes goal probability from each zone?



Result 2: Quality-Quantity Trade-Offs

Critical Insight: Increasing action frequency reduces success rates



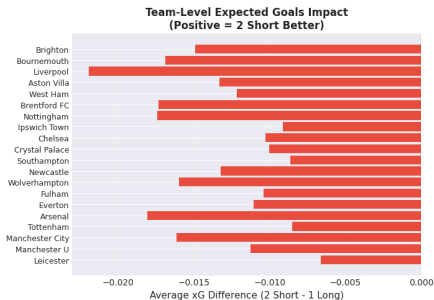
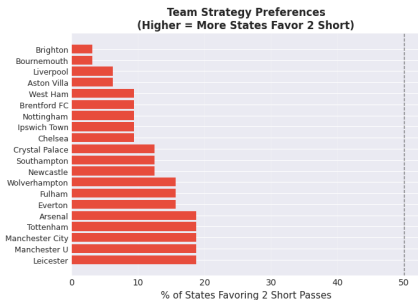
Result 3: Counterfactual Policy Analysis

Research Question: What if teams altered their pass policies?

Goal Difference from Current Policy (Goals/Season)													
Question 1: Forward Passes (short + long forward in columns 6-9)							Question 2: Long Passes (backward + lateral + forward in columns 6-9)						
	-20%	-10%	-5%	+5%	+10%	+20%		-20%	-10%	-5%	+5%	+10%	+20%
Liverpool	8.6	4.4	2.2	-2.3	-4.6	-9.3	Liverpool	-1.7	-0.9	-0.4	0.5	0.9	1.9
Brentford FC	8.2	4.2	2.1	-2.1	-4.2	-8.5	Brentford FC	-1.6	-0.8	-0.4	0.4	0.8	1.7
Manchester City	6.3	3.3	1.7	-1.7	-3.4	-7.0	Manchester City	-0.3	-0.2	-0.1	0.1	0.2	0.5
Brighton	7.0	3.6	1.8	-1.8	-3.7	-7.5	Brighton	-1.7	-0.9	-0.4	0.5	0.9	1.9
Newcastle	6.8	3.5	1.7	-1.8	-3.6	-7.2	Newcastle	-1.4	-0.7	-0.4	0.4	0.8	1.6
Nottingham	7.5	3.8	1.9	-1.9	-3.9	-7.9	Nottingham	-1.5	-0.8	-0.4	0.4	0.8	1.6
Chelsea	6.3	3.2	1.6	-1.7	-3.3	-6.8	Chelsea	-0.8	-0.4	-0.2	0.2	0.5	1.0
Tottenham	6.6	3.4	1.7	-1.7	-3.5	-7.0	Tottenham	-1.3	-0.7	-0.3	0.3	0.7	1.4
Wolverhampton	6.9	3.5	1.8	-1.8	-3.7	-7.4	Wolverhampton	-1.1	-0.6	-0.3	0.3	0.6	1.3
Arsenal	5.9	3.0	1.5	-1.6	-3.2	-6.4	Arsenal	-0.4	-0.2	-0.1	0.1	0.2	0.5
Aston Villa	5.2	2.7	1.3	-1.4	-2.8	-5.7	Aston Villa	-0.9	-0.5	-0.2	0.2	0.5	1.0
Fulham	6.1	3.1	1.6	-1.6	-3.2	-6.5	Fulham	-1.2	-0.6	-0.3	0.3	0.7	1.4
Bournemouth	5.4	2.8	1.4	-1.4	-2.9	-5.8	Bournemouth	-1.7	-0.9	-0.4	0.4	0.9	1.9
West Ham	5.4	2.8	1.4	-1.4	-2.8	-5.7	West Ham	-1.2	-0.6	-0.3	0.3	0.6	1.3
Crystal Palace	5.4	2.8	1.4	-1.4	-2.8	-5.7	Crystal Palace	-0.6	-0.3	-0.2	0.2	0.3	0.7
Everton	5.3	2.7	1.4	-1.4	-2.8	-5.6	Everton	-1.3	-0.6	-0.3	0.3	0.7	1.4
Manchester U	5.3	2.7	1.4	-1.4	-2.8	-5.6	Manchester U	-1.1	-0.6	-0.3	0.3	0.6	1.2
Ipswich Town	4.4	2.2	1.1	-1.1	-2.3	-4.6	Ipswich Town	-0.9	-0.4	-0.2	0.2	0.5	0.9
Leicester	5.2	2.6	1.3	-1.3	-2.7	-5.3	Leicester	-0.6	-0.3	-0.2	0.2	0.3	0.7
Southampton	4.5	2.3	1.1	-1.2	-2.3	-4.7	Southampton	-0.6	-0.3	-0.2	0.2	0.3	0.7

Result 4: Directness vs. Patient Build-Up

Research Question: 1 long forward pass vs 2 short forward passes in attacking half?



Key Findings (Monte Carlo simulation, 3000 runs):

- **All 20 teams** benefit from directness in 80-97% of valid positions
- League average: **+0.0132 xG** for 1 long forward pass
- Result holds across all playing styles

Summary & Contributions

What We Did:

- 1 Extended Van Roy's shooting MDP to **pass policy analysis**
- 2 Monte Carlo simulation for strategy comparison
- 3 Otherwise similar methodology

Practical Applications:

- Tactical coaching: Which zones favor long vs short passes?
- Match preparation: Opponent analysis
- Recruitment: Identify decision-making quality

Key Findings:

- Teams benefit from longer passes
- Teams benefit from less forward passes
- Team-specific optimal policies

Limitations:

- Grid discretization (7×11) loses detail
- Markov property assumption
- No opponent modeling
- Absolute xG values overestimated

Future Work:

- Finer grid resolution
- Player-level policies
- Opponent modeling

Thank You!

Questions?